1. unionSets(ArrayList<Integer> listA, ArrayList<Integer> listB)
    a. Test Case 1: the same set: {1, 2}
        i. Expected Output: {1, 2}
        ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1,2));

        x.unionSets(listA, listB);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\User
The union of set A and set B is [1, 2]
```

    b. Test Case 2: overlapping sets: {1, 2, 3, 4}, {3, 4, 5, 6}
        i. Expected Output: {1,2, 3, 4, 5, 6}
        ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2, 3, 4));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));

        x.intersectSets(listA, listB);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.
The intersection of set A and set B is [3, 4]
```

2. intersectSets(ArrayList<Integer> listA, ArrayList<Integer> listB)
    a. Test Case 1: no overlap
        i. Expected Output: {}
        ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));

        x.intersectSets(listA, listB);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sc
The intersection of set A and set B is []
```

b. Test Case 2: overlapping sets: {1, 2, 3, 4}, {3, 4, 5, 6}
    i. Expected Output: {3, 4}
    ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1,2, 3, 4));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));

        x.intersectSets(listA, listB);
```

    iii. Output:

3. findComplement(ArrayList<Integer> universalList, ArrayList<Integer> listA)
    a. Test Case 1: setA: {1, 2, 3, 4}, Universal set: {1, 2, 3, 4, 5, 6, 7}
        i. Expected Output: {5, 6, 7}
        ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
        ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
        x.findComplement(universalList, listA);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.
The complement of set A is [5, 6, 7]
```

4. findDifference(ArrayList<Integer> listA, ArrayList<Integer> listB)
    a. Test Case 1: setA: {1, 2, 3, 4}, Universal set: {3, 4, 5, 6}
        i. Expected Output: A – B: {1, 2}, B – A: {5, 6}
        ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
        ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
        x.findDifference(listA, listB);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\soph
The difference of set A minus set B is [1, 2]
The difference of set B minus set A is [5, 6]
```

5. findSymmetricDifference(ArrayList<Integer> listA, ArrayList<Integer> listB)
    a. Test Case 1: Set A: {1, 2, 3, 4}, Set B = {3, 4, 5, 6}
        i. Expected Output: {1, 2, 5, 6}
        ii. Input:

```java
public static void main(String[] args) {
    Operations x = new Operations();
    ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
    ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4, 5, 6));
    ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
    x.findSymmetricDifference(listA, listB);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\plugins\
The symmetric difference of set A and set B is [1, 2, 5, 6]
```

6. findCrossProduct(ArrayList<Integer> listA, ArrayList<Integer> listB)
    a. Test Case 1: Set A: {1, 2}, Set B = {3, 4}
        i. Expected Output: {(1, 3), (1, 4), (2, 3), (2, 4)}
        ii. Input:

```java
public static void main(String[] args) {
    Operations x = new Operations();
    ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2));
    ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
    ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
    x.findCrossProduct(listA, listB);
```

        iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\plugins\org.eclipse.justj.op
The cross product of set A and set B is [(1, 3), (1, 4), (2, 3), (2, 4)]
```

7. getPowerSet(ArrayList<Integer> listA)
    a. Test Case 2: (1, 2, 3)
        i. Expected Output: {{}, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}}
        ii. Input:

```
public static void main(String[] args) {
    Operations x = new Operations();
    ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
    ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
    ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
    x.getPowerSet(listA);
```

iii. Output:

```
Problems  Javadoc  Declaration  Console
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0
The power set of set A is [[], [1], [2], [1, 2], [3], [1, 3], [2, 3], [1, 2, 3]]
```

8. checkIfSubset(ArrayList<Integer> possibleSubset, ArrayList<Integer> possibleSuperset)
    a. Test Case 1: Is not a subset
        i. Expected Not a subset
        ii. Input:

```
public static void main(String[] args) {
    Operations x = new Operations();
    ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
    ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(3, 4));
    ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
    x.checkIfSubset(listA, listB);
```

        iii. Output:

```
Problems  Javadoc  Declaration  Console
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\pl
The first list is not a subset of the second list
```

    b. Test Case 2: Is a subset
        i. Expected Is a subset
        ii. Input:

```
public static void main(String[] args) {
    Operations x = new Operations();
    ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
    ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
    ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
    x.checkIfSubset(listA, listB);
```

        iii. Output:

```
Problems  Javadoc  Declaration  Console
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2
The first list is a subset of the second list
```

9. checkIfProperSubset(ArrayList<Integer> possibleSubset, ArrayList<Integer> possibleSuperset)
    a. Test Case 1: Is a proper subset (same sets)
        i. Expected Not a proper subset

ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1, 2, 3));
        ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
        x.checkIfProperSubset(listA, listB);
```

iii. Output:

Problems @ Javadoc Declaration Console ×

<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\plugins\

```
The first list is not a proper subset of the second list
```

b. Test Case 2: Is a proper subset

i. Expected Is a proper subset

ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(1, 2, 3, 4, 5));
        ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
        x.checkIfProperSubset(listA, listB);
```

iii. Output:

Problems @ Javadoc Declaration Console ×

<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\plugins

```
The first list is a proper subset of the second list
```

10. checkIfDisjoint(ArrayList<Integer> listA, ArrayList<Integer> listB)

a. Test Case 1: are disjoint

i. Expected Sets are disjoint

ii. Input:

```java
    public static void main(String[] args) {
        Operations x = new Operations();
        ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3));
        ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(4, 5, 6));
        ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
        x.checkIfDisjoint(listA, listB);
```

iii. Output:

Problems @ Javadoc Declaration Console ×

<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p

```
The sets are disjoint
```

b. Test Case 2: Are not disjoint

i. Expected Subsets are not disjoint

ii. Input:

```java

public static void main(String[] args) {
    Operations x = new Operations();
    ArrayList<Integer> listA = new ArrayList<>(Arrays.asList(1, 2, 3, 4));
    ArrayList<Integer> listB = new ArrayList<>(Arrays.asList(4, 5, 6 ));
    ArrayList<Integer> universalList = new ArrayList<>(Arrays.asList(1,2,3,4,5,6,7));
    System.out.println("The sets are not disjoint\n\n");
    x.checkIfDisjoint(listA, listB);

```

iii. Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> SetOperationsTester [Java Application] C:\Users\sophi\.p2\pool\
The sets are not disjoint
```